ChatGPT

Airwriting Detection: Concise Next-Step Plan

Executive summary: Audit and standardize the PNG–video–JSON sample triplets, establish writer-independent baselines, then test a cross-view masked transformer under unseen-user and personalization evaluations.

Immediate preprocessing

- **Inventory:** count `.json`, `.png`, `.webm`, and `.mp4` files by user and label.
- **Build** `manifest.csv`: `sample_id,user_id,label,json_path,image_path,video_path,device,created_at`.
- **Match triplets:** require one JSON, one trajectory PNG, and preferably one video per sample; the repository intentionally stores these under `output/{user}/{label}/`.
- **Validate JSON:** required fields: `sample_id`, `username`, `label`, `frame_size`, `trajectory`, `user_agent`.
- **Validate trajectory:** finite values, coordinates within `[0,1]`, matching `num_points`, `stroke_ids`, and `features_8d`.
- **Normalize labels:** Unicode NFC; confirm the 11 Bangla vowels and detect path/JSON label mismatches.
- **Preserve groups:** never split samples from the same user across train and test.
- **Record exclusions:** missing files, unreadable media, duplicate sample IDs, corrupt JSON, or fewer than 10 valid points.

Cleaning and augmentation

- **Cleaning:** remove NaN/Inf points, zero-length samples, repeated consecutive points, severe tracking jumps, and exact/near duplicates using file hashes plus trajectory DTW distance.
- **Trajectory normalization:** center on bounding-box centroid; scale by maximum side; preserve aspect ratio; retain stroke-boundary tokens.
- **Resampling:** interpolate each complete trajectory to **128 points**; also retain the original variable-length sequence.
- **Class balance:** use weighted sampling; avoid copying minority-user samples into another evaluation fold.
- **Trajectory augmentation:** rotation **±7°**, scale **0.90–1.10**, translation **±5%**, Gaussian jitter **σ=0.003–0.008**, point dropout **3–8%**, temporal warp **0.8–1.25×**.
- **Video augmentation:** brightness/contrast **±15%**, motion blur kernel **3–5**, frame dropout **≤10%**, crop/translation **≤5%**.
- **Keypoint augmentation:** coordinate jitter **σ=0.005**, temporal masking **5–15%**, small global rotation **±7°**.
- Apply augmentation **only after user-level splitting**.

Features and modeling

- **Raw sequence:** existing normalized `(x,y)`, stroke IDs, and repository-provided 8-D vectors: displacement, direction, curvature, and stroke-transition indicators.
- **Derived dynamics:** velocity, acceleration, curvature, cumulative path length, dwell time, stroke duration, and inter-stroke gap.
- **Time-series baseline:** masked **BiGRU**, **TCN**, or small Transformer over `[T,D]`.
- **Spatial baseline:** convert PNG to a **224×224 one-channel** trajectory image; test a small ResNet or EfficientNet.
- **Skeleton stream:** re-extract **21 hand landmarks** `(x,y,z)` **per frame** from video; wrist-center and normalize by palm size.
- **Skeleton model:** ST-GCN or temporal Transformer over `[T,21,3]`.
- **Required baselines:** logistic regression/XGBoost on summary features, 1-D CNN on sequences, and CNN on trajectory images.

New methodology

- **Cross-View Stroke-Masked Transformer:** jointly encode the 8-D trajectory, rasterized PNG, and re-extracted hand skeleton while masking trajectory segments and enforcing cross-view contrastive consistency, allowing the model to learn writer-invariant structure from all three repository outputs rather than treating each modality separately.

Evaluation protocol

- **Primary split:** leave-one-user-out cross-validation when at least five users exist.
- **Fallback:** grouped five-fold cross-validation, reducing folds to the number of available users.
- **Model selection:** grouped three-fold inner validation; test users remain completely untouched.
- **Primary metric:** macro-F1.
- **Also report:** balanced accuracy, per-class precision/recall/F1, confusion matrix, top-2 accuracy, and ROC-AUC one-vs-rest.
- **Reliability:** expected calibration error and accuracy–coverage curves with an abstention threshold.
- **Efficiency:** median inference latency, parameter count, and peak memory.
- **User-centric tests:** unseen-user zero-shot accuracy; **1-, 3-, and 5-shot** personalization; cross-device/browser; left- versus right-hand; poor-lighting subsets.
- **Ablations:** sequence only, image only, skeleton only, multimodal fusion, no augmentation, and no masked pretraining.
- Report mean, standard deviation, and **95% bootstrap confidence intervals by user**, not merely by sample.

Tooling and starter commands

```
pip install numpy pandas scipy scikit-learn torch torchvision \
  opencv-python mediapipe albumentations tslearn
```

```
find output -type f | awk -F. '{print tolower($NF)}' | sort | uniq -c
```

```
find output -name '*.json' -print0 |
xargs -0 -n1 jq -e \
'.sample_id and .username and .label and
 .trajectory.points_normalized and .trajectory.features_8d'
```

```
find output -type f \( -name '*.webm' -o -name '*.mp4' \) -print0 |
xargs -0 -n1 ffprobe -v error -show_entries \
format=duration:stream=codec_name,width,height,r_frame_rate -of json
```

- First output files: `manifest.csv`, `excluded_samples.csv`, `class_user_counts.csv`, and fixed user-level fold assignments.

Timeline

- **Days 1–2 — Data readiness:** inventory, manifest, format validation, duplicate detection, cleaning, and frozen user splits.
- **Days 3–5 — Reproducible baselines:** summary-feature, sequence, trajectory-image, and skeleton models with the full evaluation script.
- **Days 6–10 — Proposed method:** cross-view masked model, multimodal ablations, unseen-user tests, few-shot personalization, confidence intervals, and final result tables.
